

Raymond Yuan

Framingham, MA | (508) 309-2865 | yuanqingxiao@gmail.com

LinkedIn: <https://www.linkedin.com/in/raymondqxyuan>

GitHub: <https://github.com/osuyuanqi>

Professional Summary

Senior Software Engineer focused on reliability engineering and system performance. Builds deterministic validation pipelines for AI-assisted workflows and optimizes large-scale infrastructure. Proven track record of reducing resource consumption by **60%** and improving system stability by **16%** through data-driven analysis.

Experience

The MathWorks | Senior Software Engineer in Test

Sep 2022 – Present

- **Used LLM tools (e.g., Claude) to assist infrastructure analysis**, accelerating identification of resource-heavy workflows and driving **60%+ memory reduction**.
- **Reduced system-wide flakiness by 16%** across core features through systematic root cause analysis and infrastructure hardening.
- **Defined and implemented reliability improvements** for infrastructure used across multiple product teams, improving stability and test consistency.
- **Led performance profiling of core infrastructure**, identifying bottlenecks and improving execution stability across regression systems.
- **Applied validation and fallback patterns to AI-assisted workflows** to prevent regressions in test systems and ensure deterministic outputs.
- **Evaluated and validated AI-assisted outputs**, implementing deterministic fallbacks and safeguards to ensure system correctness in mission-critical environments.
- **Drove cross-team execution** with Release Engineering to ensure stable execution across multiple release cycles.
- **Spearheaded codebase management (Perforce)** initiatives, mentoring peers to improve team-wide version control discipline.
- **Identified and resolved critical lifecycle bugs** in Simulink Real-Time and SDI, reducing production regression risks.

CrowdDoing (Match4Action) | UI/UX Designer (Volunteer)

May 2022 – Sep 2022

- **Contributed to product design discussions**, applying user psychology to enhance platform usability and accessibility.

- **Professional Pivot:** Leveraged design insights to strengthen the intersection of UX and performance testing.

Sublime China Information Co., Ltd. (SCI) | Full-Stack Software Developer

Sep 2012 – Feb 2018

- **Led a 3-engineer team** building a mobile-responsive industry platform for 10,000+ daily active users.
- **Developed full-stack applications** and managed the complete SDLC from requirements to final copyright submission.
- **Built industry-standard messaging systems** and optimized data visualization tools.
- **Mentored junior engineers** as the primary lead for complex JavaScript challenges.

Featured Project

Gemini Blind Timer | Full-Stack / AI Infrastructure Integration | May 2026

GitHub: <https://github.com/osuyuanqi/random-timer>

Demo: <https://osuyuanqi.github.io/random-timer>

- **Evaluated reliability tradeoffs** between LLM-driven and deterministic randomness, optimizing for output consistency under varying latency conditions.
- **Built a robust CI/CD gate (Python + pytest) to enforce correctness and consistency** of AI-assisted outputs under API variability.
- **Implemented a Firestore persistence layer** with consistency guarantees and real-time synchronization.

Education

Oregon State University | 2019 – 2021

M.Eng., Computer Science | GPA: 3.86

Shandong University of Science and Technology | 2008 – 2012

B.M., Information Systems

Skills

- **Programming:** Python, JavaScript/Node.js, MATLAB.
- **AI & LLM:** LLM APIs (Gemini, Claude), LLM evaluation workflows, Prompt Engineering.
- **Infrastructure & Reliability:** Performance profiling, Regression analysis, Framework Hardening, System scaling.
- **Tools:** Perforce (P4), Git, Linux, GitHub Actions, SQL, Firestore.